Home Wi-Fi Network Devices

Question 1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home Wi-Fi setup.

Answer: Network devices have different roles. A switch connects wired devices within a local area network (LAN); a router forwards traffic between networks; a hub repeats incoming signals to all its ports; a firewall filters traffic according to security rules; and a wireless access point (AP) connects Wi-Fi devices to the LAN.

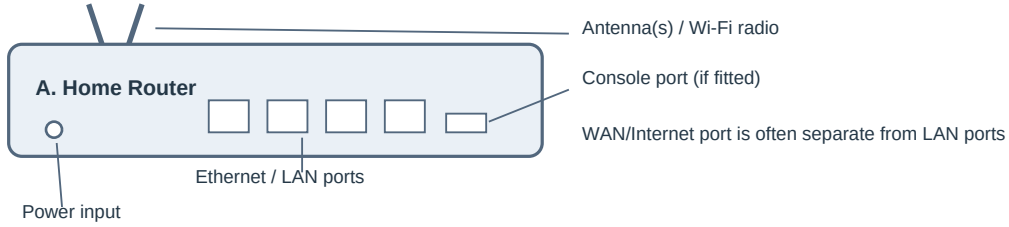
Device	Main function	Typical physical ports / interfaces	Typical network ports / protocols
Switch	Connects wired devices in the same LAN and forwards Ethernet frames to the appropriate port using MAC addresses.	Several Ethernet RJ-45 ports; sometimes an uplink/SFP port; power input. Managed models may have a console port.	No single required TCP/UDP port for normal switching. Management may use HTTPS 443, SSH 22, SNMP 161/162, or Telnet 23 if enabled.
Router	Connects different networks (for example, home LAN to ISP/WAN), routes IP packets, and often provides NAT, DHCP and basic firewall features.	WAN/Internet Ethernet port, one or more LAN Ethernet ports, power input; some models have USB or DSL/fiber interfaces.	Routing itself is not identified by one TCP/UDP port. Management may use HTTPS 443 or SSH 22; DHCP uses UDP 67/68; DNS usually uses 53.
Hub	A basic Layer 1 device that repeats incoming electrical signals to all other ports; it does not learn MAC addresses or filter frames.	Multiple Ethernet RJ-45 ports and a power input on many models.	No TCP/UDP service ports for normal hub operation; it repeats physical signals rather than inspecting application traffic.
Firewall	Allows or blocks network traffic according to rules; helps protect a network from unwanted connections. It may be a separate appliance or built into a router.	Often WAN and LAN Ethernet ports; sometimes management/console port and power input. Software firewalls have no dedicated physical ports.	No single fixed port: it filters traffic on many ports. Management can use HTTPS 443 or SSH 22; common allowed services may include DNS 53 and HTTPS 443.
Access point (AP)	Provides Wi-Fi coverage and bridges wireless clients to the wired LAN; may support multiple SSIDs and guest Wi-Fi.	Ethernet uplink (often PoE-capable), power input or PoE, and internal/external antennas. Some models have a reset button.	Wi-Fi uses radio channels, not TCP/UDP ports. Management may use HTTPS 443 or SSH 22; client traffic uses the ports of the apps and services being accessed.

Important distinction: “Physical ports” are connectors on the device (such as Ethernet, console and power). “Network ports” are logical TCP/UDP numbers used by services (such as HTTPS 443). These are different meanings of the word *port*. The values above are typical examples, not a list of ports that every device must use.

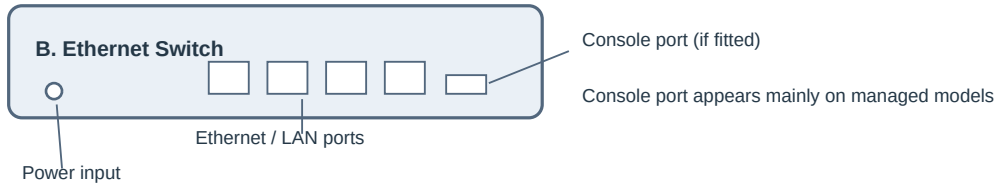
Question 2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Answer: The following simplified diagrams show common device shapes and the labels that should be marked. Actual port positions vary by manufacturer and model. Use the labels as a guide when identifying real devices.

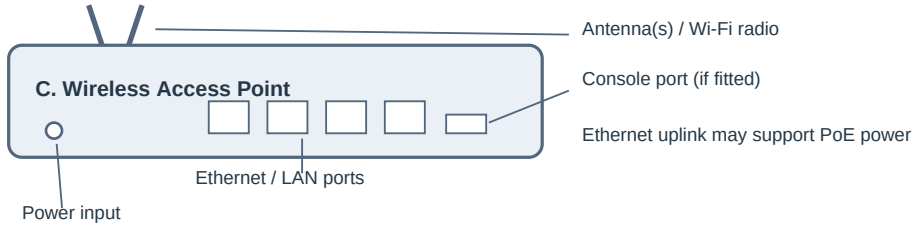
A. Home Router



B. Ethernet Switch



C. Wireless Access Point



How to label the diagrams step by step:

1. **Ethernet:** mark the rectangular RJ-45 network sockets. On a router, identify the WAN/Internet socket separately from LAN sockets if the labels are present.
2. **Console:** mark the dedicated console socket only if the model has one; it is often used for initial configuration on managed network equipment.
3. **Power:** mark the DC power jack or power connector. Some access points receive power through Ethernet using Power over Ethernet (PoE), so a separate power jack may not be used.
4. **Antenna:** mark external antenna rods if present. Many modern routers and APs use internal antennas, which are not visible from outside.

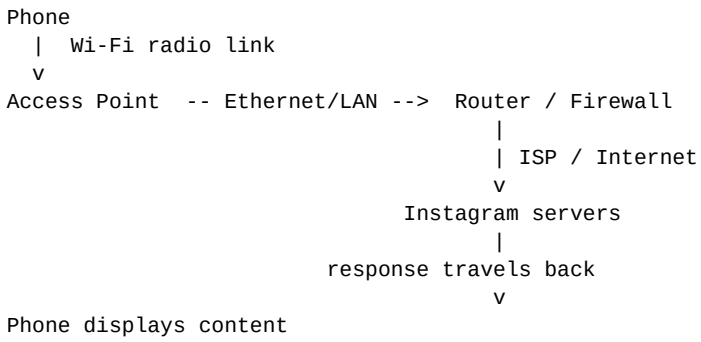
Note: These are schematic illustrations, not exact manufacturer-specific photographs. When labeling an actual image, rely on the port names printed beside the sockets and the device's model guide.

Question 3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3–4 lines how at least two different network devices play a role when you use that app.

Answer — Example: Instagram (4 lines)

1. My phone connects to the home **access point** over Wi-Fi, which carries the phone's network traffic to the home LAN.
2. The **router** forwards that traffic from the home network toward the internet and usually performs NAT so multiple home devices can share one public connection.
3. A **firewall**, often built into the router or provided by the device/network, applies traffic rules and can block unwanted connections.
4. Instagram's servers return photos, reels and messages through the internet; the replies travel back through the router and access point to my phone.

Step-by-step flow:



A home Wi-Fi router often combines several roles in one box: router, small Ethernet switch, access point, and basic firewall. Separate devices are more common in larger or managed networks.

Question 4. Suppose your college Wi-Fi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason and how upgrading one device could help.

Answer: One possible reason is that the college's access points are overloaded or do not provide enough coverage for the number of students. During busy hours, many phones and laptops may share the same AP radio capacity, causing congestion, low speeds, and dropped connections—especially in crowded classrooms or areas with weak signal.

Suggested upgrade: replace or add suitable access points.

1. **Check the problem area:** note whether disconnects occur in one classroom, across an entire floor, or throughout the campus.
2. **Check coverage and capacity:** ask the network administrator to review signal strength, connected-client count, channel interference and AP load.
3. **Upgrade or add APs:** modern, correctly placed access points can improve coverage and support more simultaneous clients. Configure channels and transmit power carefully so nearby APs do not interfere with one another.
4. **Test after the change:** compare speed, signal strength and disconnections in the same locations and at similar busy times.

How the other devices can matter: A switch with insufficient capacity or a faulty uplink can bottleneck several APs. A router that is overloaded or has a limited WAN connection can slow the whole campus. If the issue affects every AP and every user, the administrator should check the router, internet link and switch uplinks before buying more APs.

Conclusion: If the problem is limited to crowded or weak-signal areas, improving access-point coverage and capacity is a reasonable first option. If the entire college network is slow, diagnose the switch uplinks, router and ISP connection as well; upgrading an AP will not fix every possible cause.

Quick revision: Switch = connects wired devices in a LAN; Router = connects networks; Hub = repeats signals to all ports; Firewall = filters traffic; Access point = provides Wi-Fi connectivity.